IoT MINI PROJECT · PHASE 1: PROJECT PROPOSAL

Addressing Class Imbalance in CICIoT2023-Based Intrusion Detection via SMOTE and Feature Selection



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Base paper: E. C. P. Neto et al., “CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment,” Sensors 23(13):5941, 2023

High accuracy can still miss the attacks that matter

≈ 2,600 : 1

DDoS flows for every Brute-Force flow in CICIoT2023

A model that always answers "DDoS" is already ≈ 73% accurate, while catching zero brute-force or web attacks.



Rare is not harmless

Brute-force logins and web injections are how IoT devices get taken over, not just knocked offline.



Accuracy hides the failure

DDoS and DoS make up ≈ 90% of the flows, so they dominate accuracy and weighted scores.



Per-class recall is the honest test

This project measures and improves recall on the rare classes, not just the headline number.

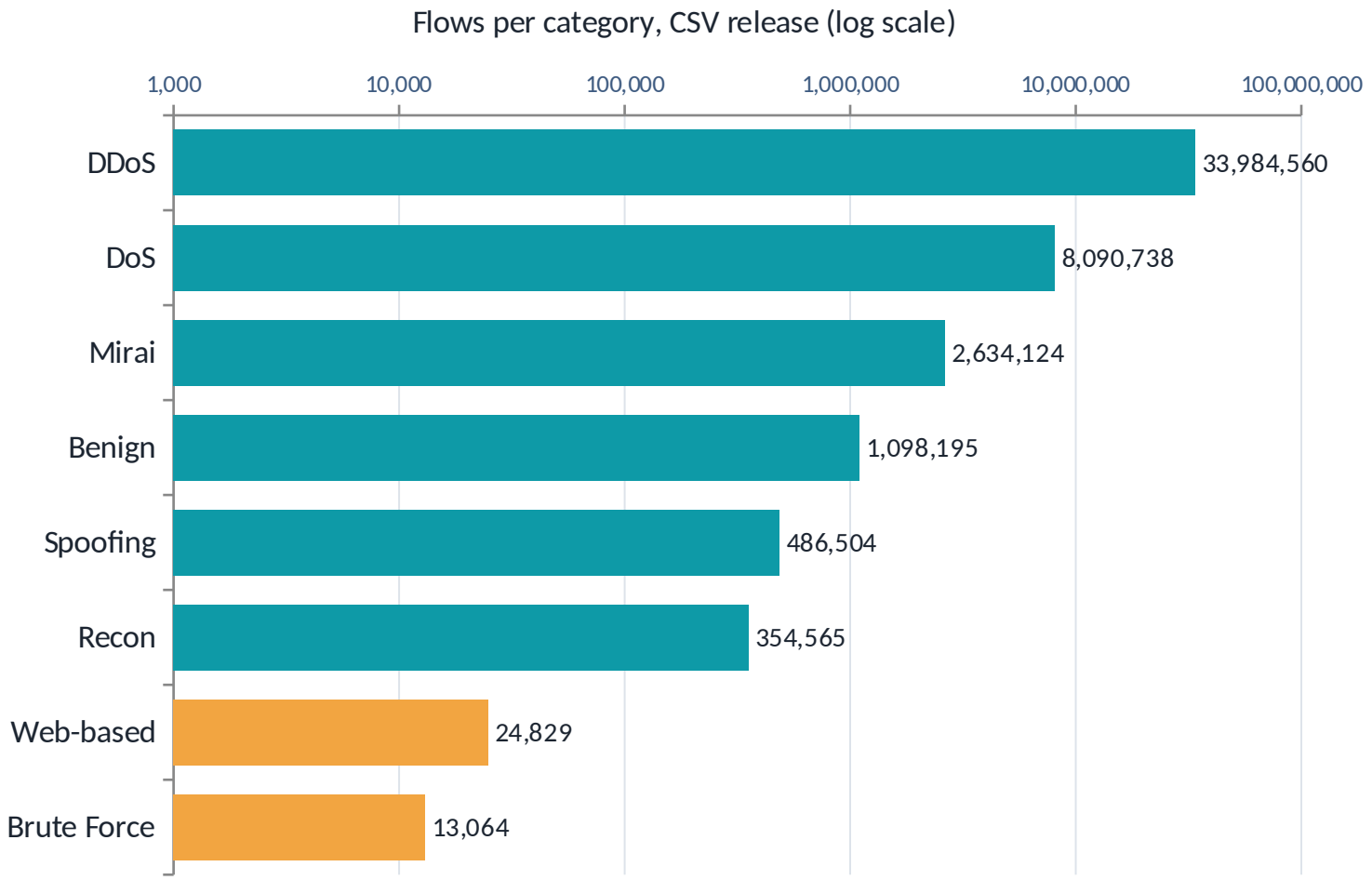
CICIoT2023: the benchmark this project builds on

105	33	7	46	≈ 46.7 M
real IoT devices	attacks	attack categories	flow features	labelled flows

Neto et al., Sensors 2023 (Canadian Institute for Cybersecurity, UNB)

- Real traffic captured in a lab topology of smart-home and IoT devices, attacked by malicious IoT devices.
- Categories: DDoS, DoS, Mirai, Reconnaissance, Spoofing, Web-based, Brute Force, plus benign traffic.
- Benchmarks machine-learning and deep-learning models at 2-class, 8-class and 34-class granularity.
- Listed in our course brief under “IoT Analytics Using Published IoT Datasets”.

The gap: two attack categories are under 0.1% of the data



Web + Brute Force

≈ 38,000 of ≈ 46.7 million flows (0.08%). A model trained on this data sees almost no examples of them.

The question

Can we improve detection of these rare attacks without collecting new data, and without hurting the common ones?

Three research questions, each with a measurable target

RQ1 · Reproduce

REPRODUCTION

Does a tree-model reproduction (Random Forest, plus Decision Tree, LightGBM and Balanced Random Forest) match the paper's CICIoT2023 benchmark?

Target: results in the paper's range, with every gap explained.

RQ2 · Rebalance

IMPROVEMENT

Does handling the imbalance (SMOTE, Borderline-SMOTE, SMOTE + undersampling, class weights, threshold tuning) raise recall on the rare classes?

Target: higher rare-class recall and macro-F1 than the baseline on the same test set, confirmed by significance tests.

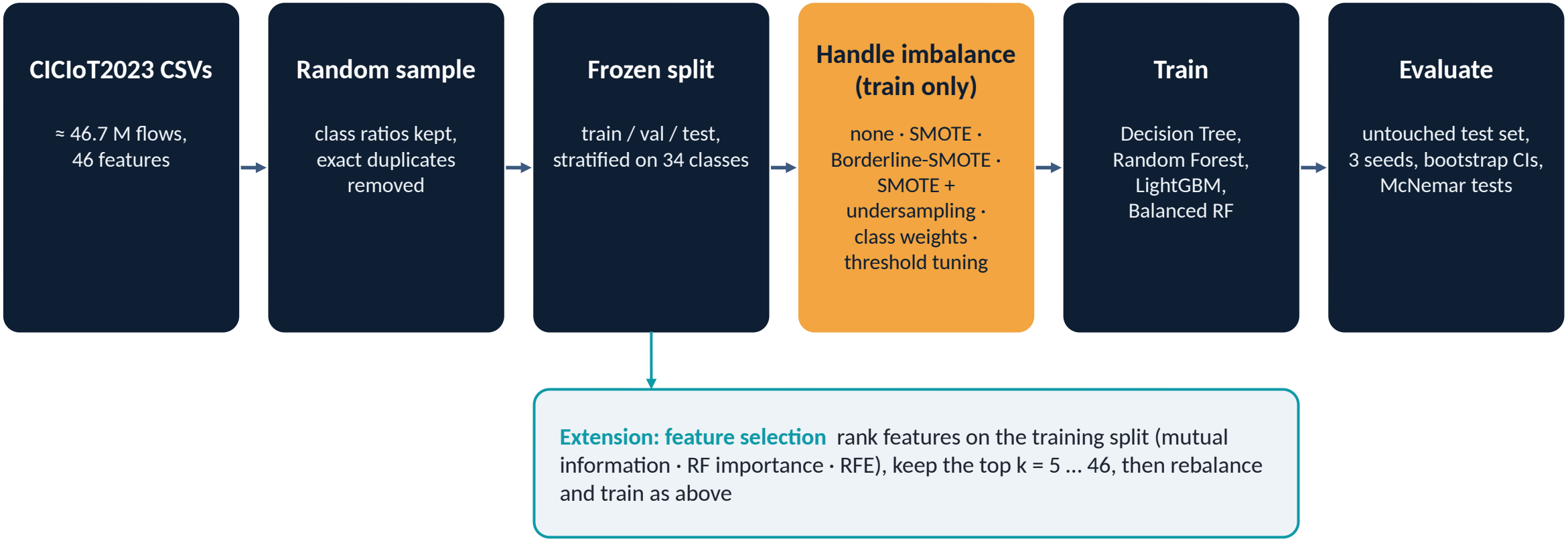
RQ3 · Reduce

EXTENSION

Can a small feature subset (mutual information, RFE) keep that gain at lower cost?

Target: ≤ 15 of 46 features keep most of the gain, with lower latency and model size.

Methodology: one pipeline, six ways to handle imbalance



Leakage guard-rails: the test set is split off first and never resampled; scaling, SMOTE and feature ranking are fit on training data only; settings are chosen on the validation split.

What this project adds beyond the base paper

	Base paper (benchmark)	This project
Headline metric	Overall accuracy and averaged scores	✓ Per-class recall and macro-F1, rare classes reported separately
Class imbalance	Imbalanced data used as-is	✓ Five imbalance methods compared head-to-head with the baseline
Models	Classical ML and deep-learning baselines	✓ Tree models, including LightGBM and Balanced Random Forest
Feature set	All 46 features	✓ Ranked subsets (MI, RF importance, RFE), from 5 to 46 features
Deployment cost	Not measured	✓ Latency per flow, model size and training time for every model
Rigour	Benchmark scores reported	✓ Duplicates removed, 3 seeds, bootstrap CIs, McNemar tests

How success will be measured

Macro-F1

Every class counts equally, so rare classes can’t hide.

Per-class recall

Share of each attack type actually caught; the core result.

Rare-class recall

Mean recall over classes below 1% of the training data.

Balanced accuracy, MCC

Imbalance-robust overall scores.

Prediction latency

Microseconds per flow, for gateway deployment.

Model size

Megabytes on disk, and tree node count.

Fair comparison: every model is scored on the same frozen, never-resampled, duplicate-free test set; mean ± standard deviation over 3 random seeds; every gain checked against the baseline with paired bootstrap 95% confidence intervals and McNemar tests.

Feasible and already de-risked

Built

pipeline implemented and tested end-to-end

< 12 h

full experiment run in one free Kaggle CPU session

₹ 0

no special hardware, public dataset

Risk	Mitigation
Dataset too large (≈ 13 GB)	Stream the files and keep a random sample with the true class ratios.
SMOTE runs out of memory	Raise rare classes to a capped target, not to the DDoS count; hybrid undersampling.
Plain SMOTE gives no gain	Four more methods (Borderline-SMOTE, hybrid, class weights, threshold tuning) are compared.
Numbers differ from the paper	Expected with a sample; the key comparison is baseline vs rebalanced on the same split.

Fits the course brief directly

Project topic 4: IoT Intrusion Detection System

- DoS: DoS and DDoS categories
- Port scanning: Reconnaissance category
- Unauthorized access: Brute Force category
- Replay attacks are not in CICIoT2023; noted as a limitation

Reading list: IoT Analytics Using Published IoT Datasets

- Base paper is item 3 (Neto et al., CICIoT2023)
- Related work draws on the security-framework list: MQTT IDS with deep learning and with LightGBM

Links to the gateway and edge topics

- The extension targets models small and fast enough for an IoT gateway or edge AI server (topics 3 and 10)
- Latency and model size are measured, not assumed

Plan and deliverables

1 · Reproduce

Sample, de-duplicate, split; baseline DT, RF, LightGBM, Balanced RF at 2, 8, 34 classes

2 · Improve

Five imbalance methods × three models vs baseline, 3 seeds, significance tests

3 · Extend

Feature ranking and top-k sweep: accuracy vs latency and size

4 · Report

Analysis, report, final presentation and demo

GitHub repo

code, tests, results

Report

PDF write-up

Presentation

results and demo

Kaggle notebook

one-click reproduction

Approval requested

Reproduce the CICIoT2023 benchmark, fix its blind spot on rare attacks, and make the fix light enough for an IoT gateway.

Clear problem

≈ 2,600 : 1 imbalance hides rare attacks

Measurable outcome

rare-class recall and macro-F1 vs a reproduced baseline

Low risk

pipeline built, public data, free compute

References
[1] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, A. A. Ghorbani, "CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment," Sensors, 23(13):5941, 2023. doi:10.3390/s23135941
[2] N. V. Chawla, K. W. Bowyer, L. O. Hall, W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," JAIR, 16:321–357, 2002.